Global and local CP asymmetries in $B^{\pm} \to \pi^{\pm}\pi^{+}\pi^{-}$ decays

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1. Introduction and Theory

Introduction: CP asymmetry

What is CP asymmetry?

- C (charge conjugation): particle ↔ antiparticle
- P (parity): spatial inversion
- CP violation: laws of physics not invariant under CP

  In Standard Model, arises from complex phase in CKM matrix [1].

Why is it important?

- One condition of Sakharov conditions for matter-antimatter asymmetry
- Standard Model CP violation is too small

  → Search for additional sources beyond the Standard Model

Theory: B meson Decay Channels

- **B meson:** $\boldsymbol{B^- = b\bar{u}, B^+ = \bar{b}u}$

Main decay channel involves (inclusive)

1. D^0 or $\overline{D^0}X \sim 79\%$ 2. $\ell^{\pm}\nu_{\ell}X \sim 11\%$ 3. $J/\psi(1S)K^+ \sim 0.1\%$

And so on…

- **We are looking for (charmless decay)**

1. $\pi^{\pm}\pi^{\pm}\pi^{+} \sim 1.52 \times 10^{-3}\%$

2. $K^{\pm}K^{\pm}K^{\mp} \sim 3.40 \times 10^{-3}\%$

3. $K^{\pm}\pi^{\pm}\pi^{\mp} \sim 5.10 \times 10^{-3}\%$

Reference [2]

Theory: Invariant Mass Reconstruction

Definition:

$$s = \left(\sum_i \widetilde{p_i}\right)^2$$

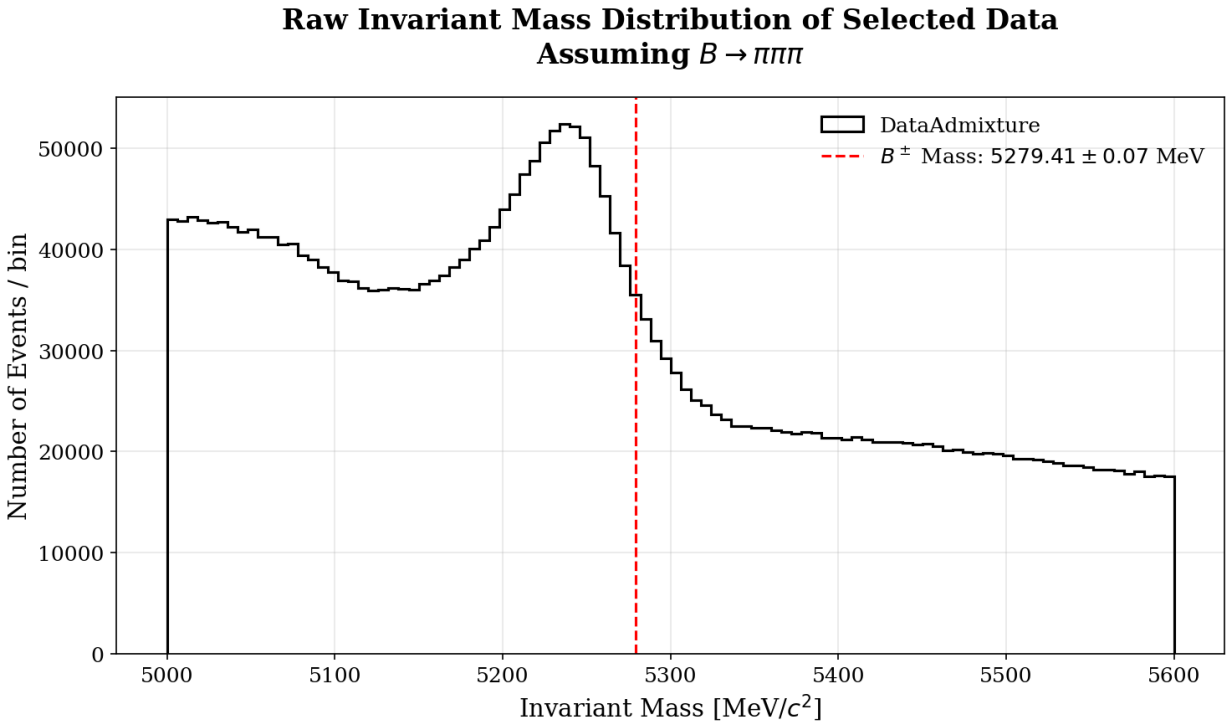
- Combine final state particle 4-momenta
- Reconstruct parent particle mass $(B^{\pm})$
- Used to identify signal over background

Theory: Invariant Mass Reconstruction

1. **Signal Overlap:** 3π signal obscured by 4-body decay background

2. **Resolution Effects:** Finite momentum resolution blurs the invariant mass shape

Observed distribution =
true shape ⊗ Gaussian resolution



2. Data Selection and Vetoing

Dataset and preselection

Dataset:

- Data from LHCb experiment
- Proton-proton collisions at $\sqrt{s} = 7\ \mathrm{TeV}$
- Forward spectrometer
  - Optimised for CP violation studies
  - Covers small angles to beam axis

[3]

Preselection:

- Applied basic cuts, including:
  - Track transverse momentum
  - Reconstructed $B^{\pm}$ mass
  - Impact parameter
- Removed muon tracks
  - Identified with muon chambers

Selection strategy overview

Goal: Maximise signal-to-noise ratio (SNR)

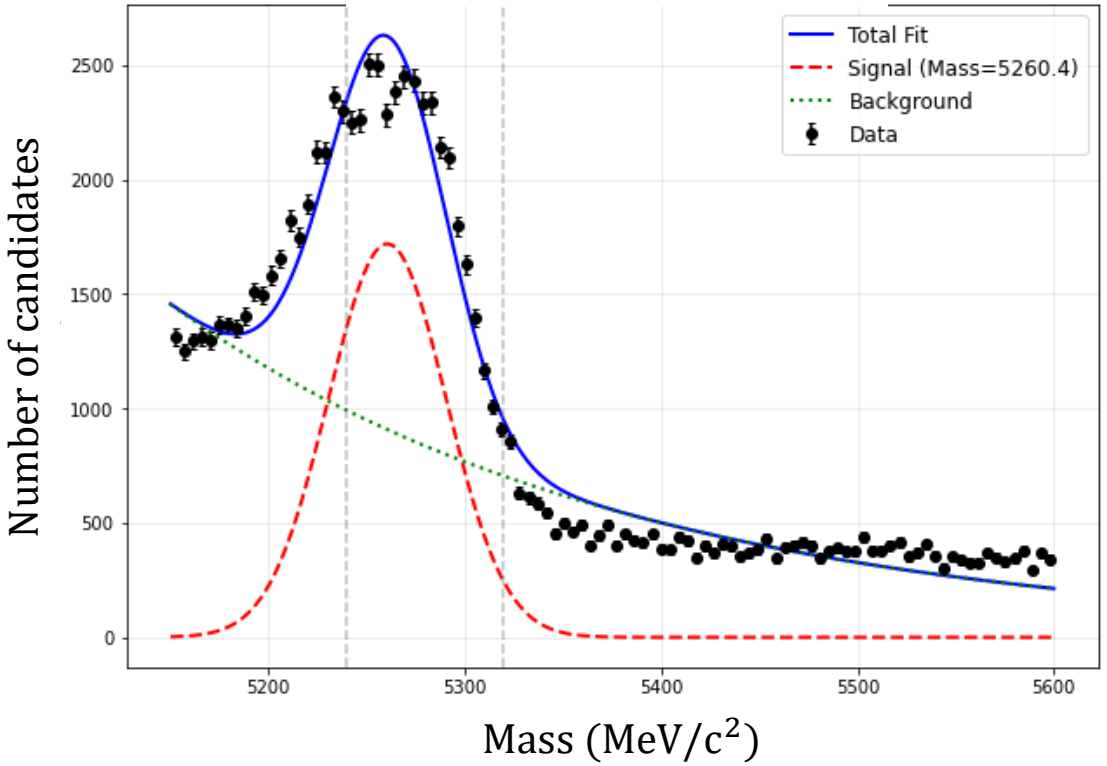
$$\frac{N_S}{\sqrt{N_S + N_B}}$$

where N_S is signal yield and N_B is background yield

Selection approach:

1. Fit invariant mass distribution at B meson peak → N_S, N_B
   *Model: Gaussian + Exponential**
2. Perform grid search over selection cuts to find best SNR

Example Fit Result



Fit model note:

Fit quality is not ideal, but has negligible impact on SNR optimisation; optimal cuts are insensitive to small modelling inaccuracies.

Selection optimisation

Beyond preselection, further cuts were explored on 3 variables:

Variable scanned	Explanation	Expectation	Best Cut
Particle ID (PID)	$P(\pi) - P(K)$	> 0: select pion-like tracks	> -0.389
Impact Parameter χ^2	How inconsistent track is with originating from primary vertex	Large: B meson travels before decaying	> 27.56
Vertex χ^2	Quality of 3-track vertex fit	Small: Good vertex quality	No cut beyond pre-cut (< 12)

Performance:

- SNR = 116.4
- Events: 5.0M -> 0.59M (12% of data kept)

D^0 $(c\bar{u})$ veto strategy

Interested in **charmless decays only**

Charm introduces different decay amplitudes and phases (CKM)

Goal: Remove data if any pair of tracks is consistent with $D^0 \to h^+h^-$

Veto method:

1. Reconstruct invariant mass for all opposite-charge track pairs under all possible particle assignments

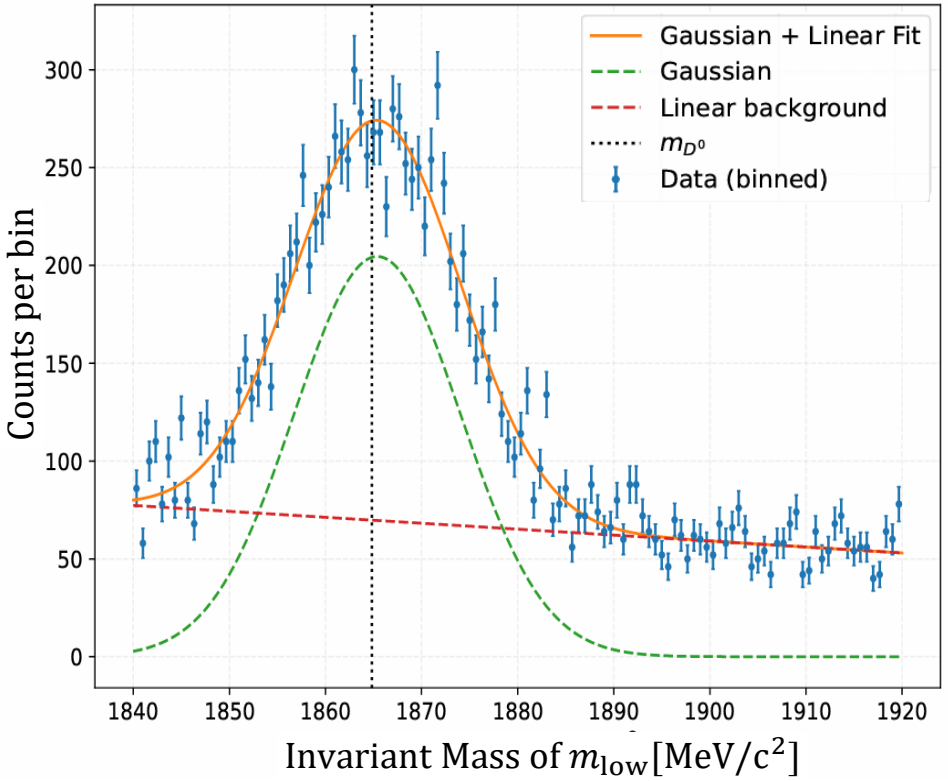
   All permutations of [2 pions + 1 kaon] / [2 kaons + 1 pions]

2. Remove peak at $m_{D^0} \approx 1.865$ GeV

   Fit Gaussian (signal) + linear background

   Apply 3 standard deviation veto around peak

Example D^0 veto fit ($\pi\pi\pi$ assumption)



J/ψ veto strategy

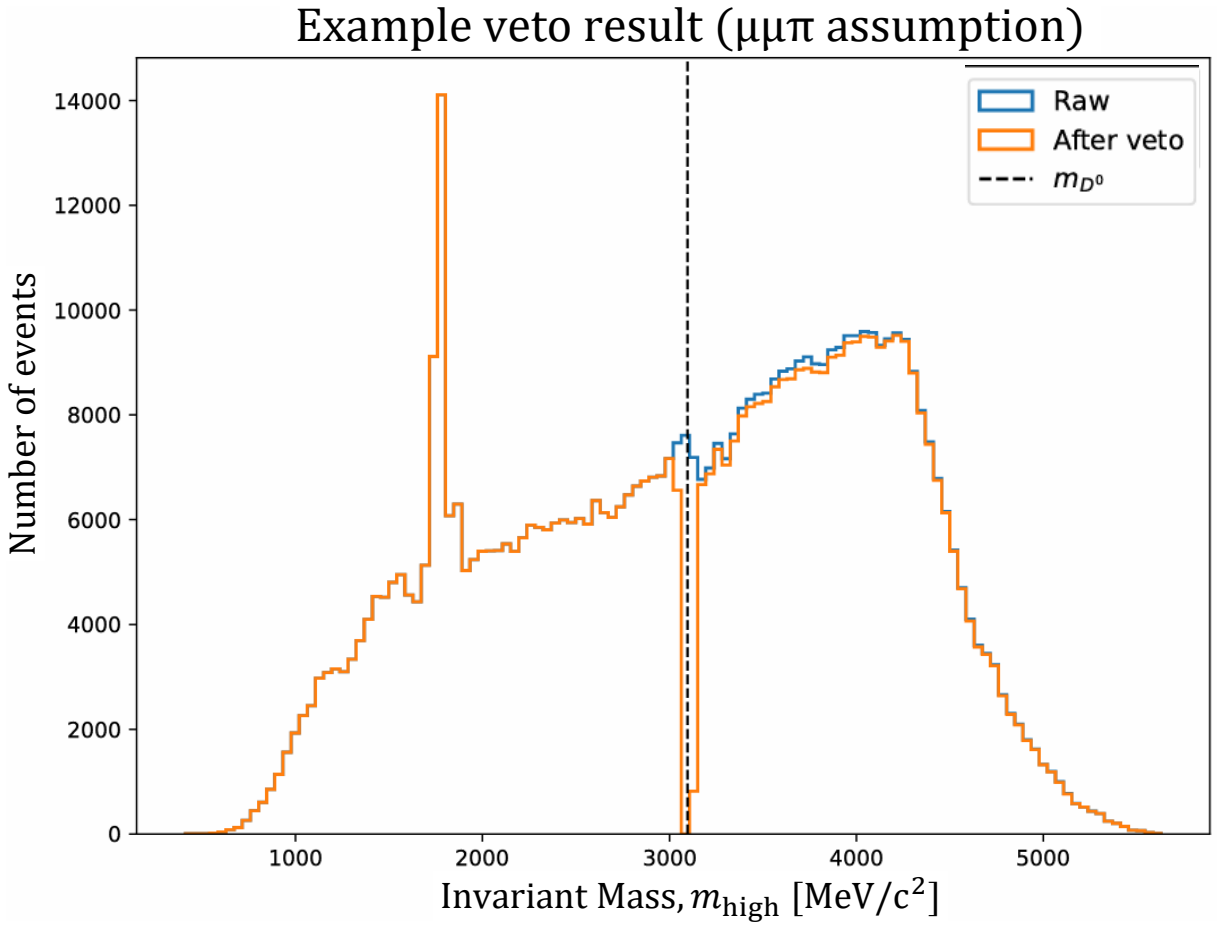
Muons may be misidentified as pions:

$$J/\psi \rightarrow \mu^+\mu^-$$

Include muon data to recover true peak and veto similarly

Veto is around:

$$m_{J/\psi} \approx 3.097 \text{ GeV}$$



Total data removed (both vetoes): $\frac{1.6\times10^5}{5.9\times10^5} = 28\%$

3. Identifying Global CP Asymmetry

Global CP asymmetry

Definition

$$A_{CP} = \frac{N^- - N^+}{N^- + N^+}$$

where N^+, N^- are signal yields for charge-conjugate decays.

Statistical uncertainty

$$\sigma_A^2 = \left(\frac{\partial A}{\partial N^-}\right)^2 \sigma_{N^-}^2 + \left(\frac{\partial A}{\partial N^+}\right)^2 \sigma_{N^+}^2 + \underbrace{2\left(\frac{\partial A}{\partial N^+}\right)\left(\frac{\partial A}{\partial N^+}\right)\mathrm{Cov}(\mathrm{N}^-, \mathrm{N}^+)}_{\textit{Cannot assume independence}}$$

Goal:

- Need precise extraction of yields by fitting
- Must separate:
  - Signal
  - Background: combinatorial background + partially reconstructed decays

Extended unbinned maximum likelihood fit

The likelihood is given by

$$\mathcal{L}_{\mathrm{ext}} = \mathrm{e}^{-\mathrm{N}} \prod_{\mathrm{i}=1} \left(\sum_{\mathrm{k}} \mathrm{N_k}\, \mathrm{f_k}(\mathrm{m_i}) \right)$$

- $\mathrm{N_k}$: yield of component k
- m_i: invariant mass of event i
- $f_k(N_k)$: PDF for component k
- First term = Poisson probability

Minimise $\boldsymbol{-2 ln \mathcal{L}_{\mathrm{ext}}}$ [7]

Fit strategy:

Fit <u>positive and negative samples together</u>

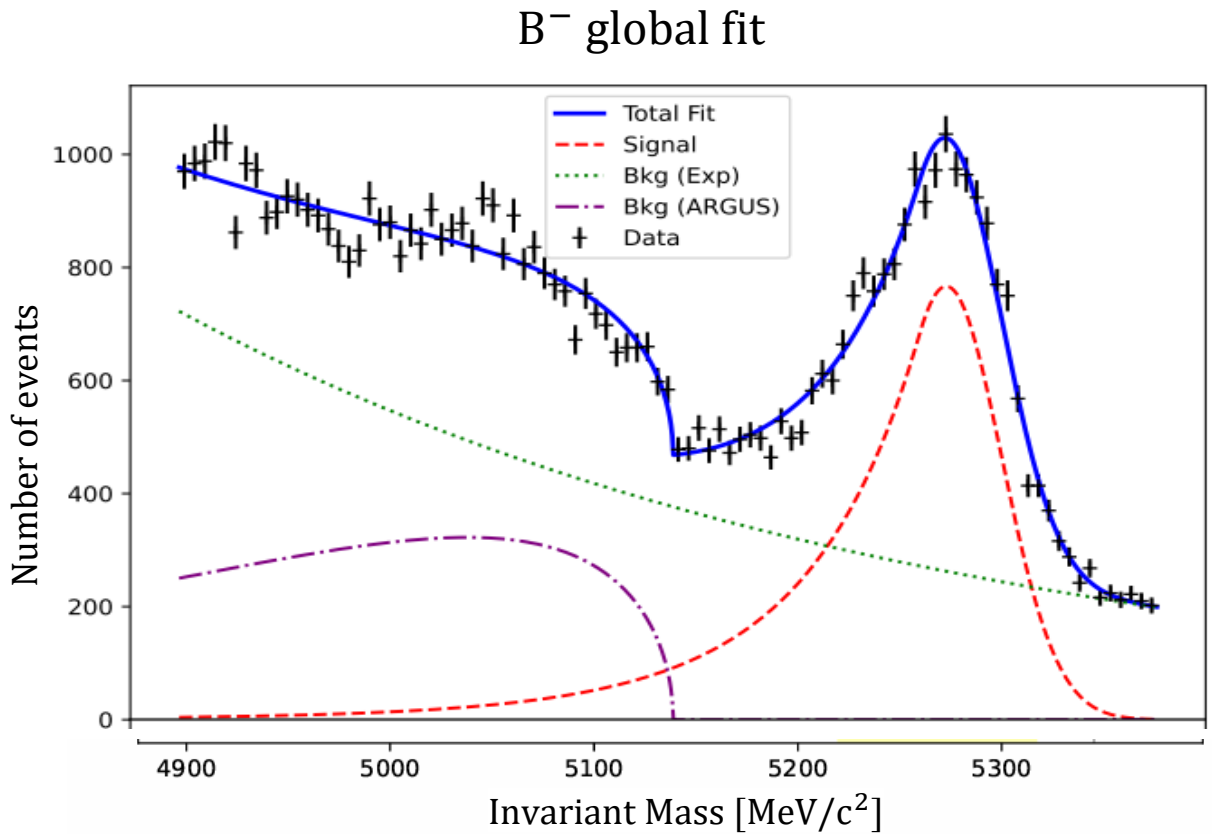
- Shared shape parameters
- Independent yields N_k^+, N_k^-

Why this approach:

- Maximises statistical precision
- Avoids binning bias
- Asymmetry from yields, not shape differences

Fit models and results

- **Signal:** Crystal ball
  - Gaussian core + power-law tail
- **Combinatorial background:** Exponential
- **Partially reconstructed:** ARGUS
  - Energy-constrained decays with unseen particles
  - Edge-like structure below $B^{\pm}$ mass



$$A_{\text{raw}} = (0.07 \pm 0.03)$$
$$A_{\text{CP (LHCb)}} = (0.076 \pm 0.008)$$

Consistent with LHCb to 1 standard deviation [2]

4. Identifying Local CP Asymmetry

What is a Dalitz plot?

- Events plotted using two body invariant masses: m_{12}^2 vs m_{13}^2

Event density:

$$\propto \left|\mathcal{A}(m_{12}^2, m_{13}^2)\right|^2$$

→ maps the decay dynamics across phase space [5]

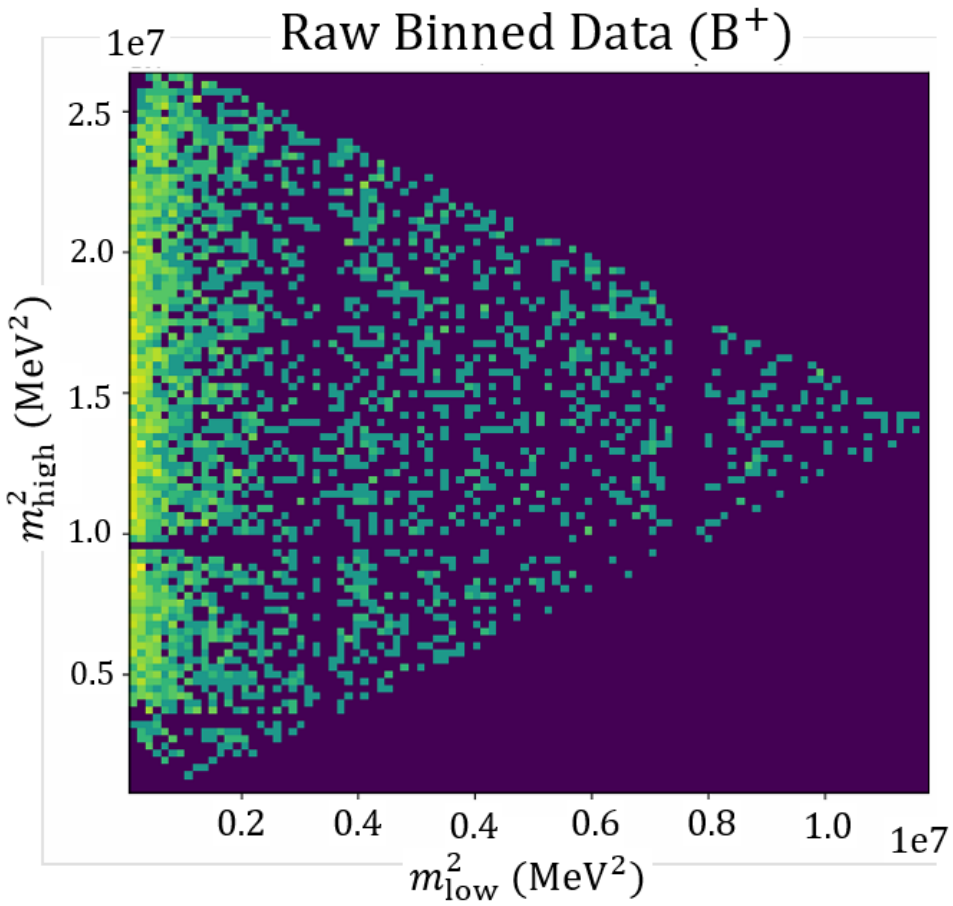
- Intermediate resonances appear as bands (fixed invariant mass)
- Multiple amplitudes → interference
- <u>CP violation arises from interference</u>
  - Due to different relative phases in B^+ vs B^-

Raw Dalitz plot

Plotting m^2_{low} vs m^2_{high} enforces symmetry

Visible features:

- Clear phase space boundary (kinematic limit)
- Large population at low invariant mass
  → likely resonance band(s)



Limitation:

- Distribution contains signal + background
- Background can hide structure and fake asymmetry

Methods for subtracting background

Two methods of subtracting background from the Dalitz plot:

sWeights

(Consistent with LHCb analyses [4])

Uses invariant mass fit to weight events by their signal probability

Sideband subtraction

Uses background-dominated regions to estimate and subtract the background

sWeights idea

Fit invariant mass m_B → PDFs $f_k(m_{B\,i})$ for each component

$$\boxed{w_i^{(\mathrm{S})}} = \sum_k \boxed{V_{Sk}} \boxed{\frac{f_k(m_{B\,i})}{\sum_j N_j f_j(m_{B\,i})}}$$

Signal weight applied to each event i

Covariance of signal yield and component k
(accounts for overlap between components)

Probability of event i belonging to component k

Weighted distribution → signal Dalitz

sWeights key assumption

Assumption:

$$m_B \perp (m_{low}^2, m_{high}^2)$$

Tests:

- Pearson, Spearman ≈ 0.07
  (Very weak linear / monotonic correlation between each Dalitz variable)

- Distance correlation ≈ 0.13 (p=0.005)
  (Weak correlation with both Dalitz variables together across a random sample)

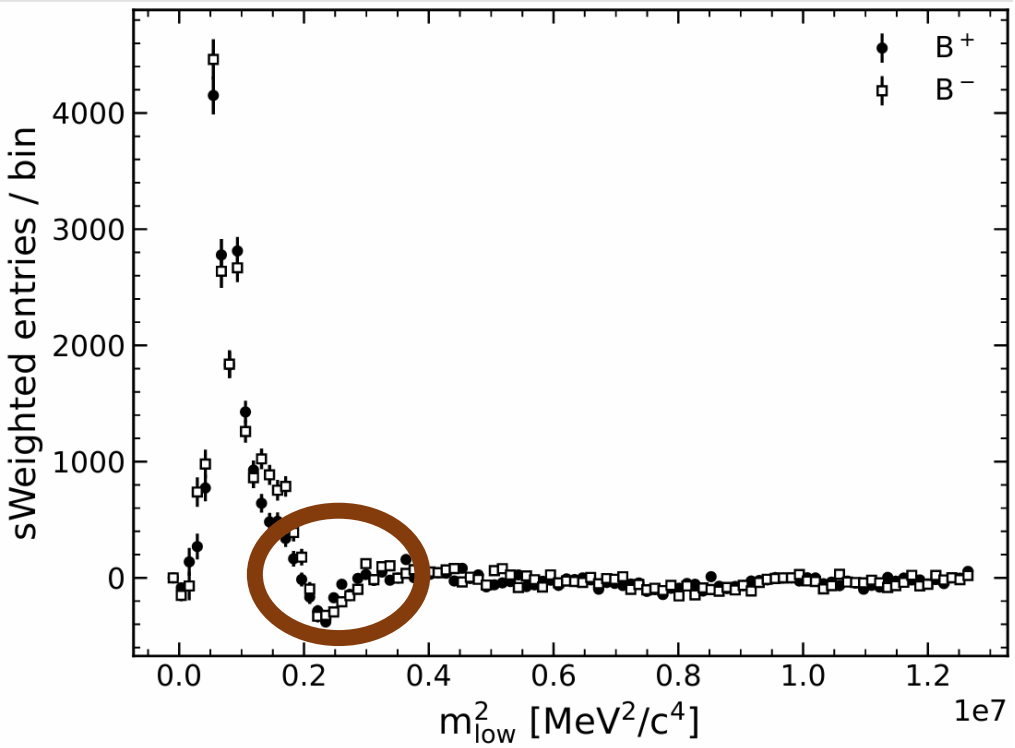
Result:
Detectable (p=0.005) but weak non-linear dependence.

Likely from kinematic boundary

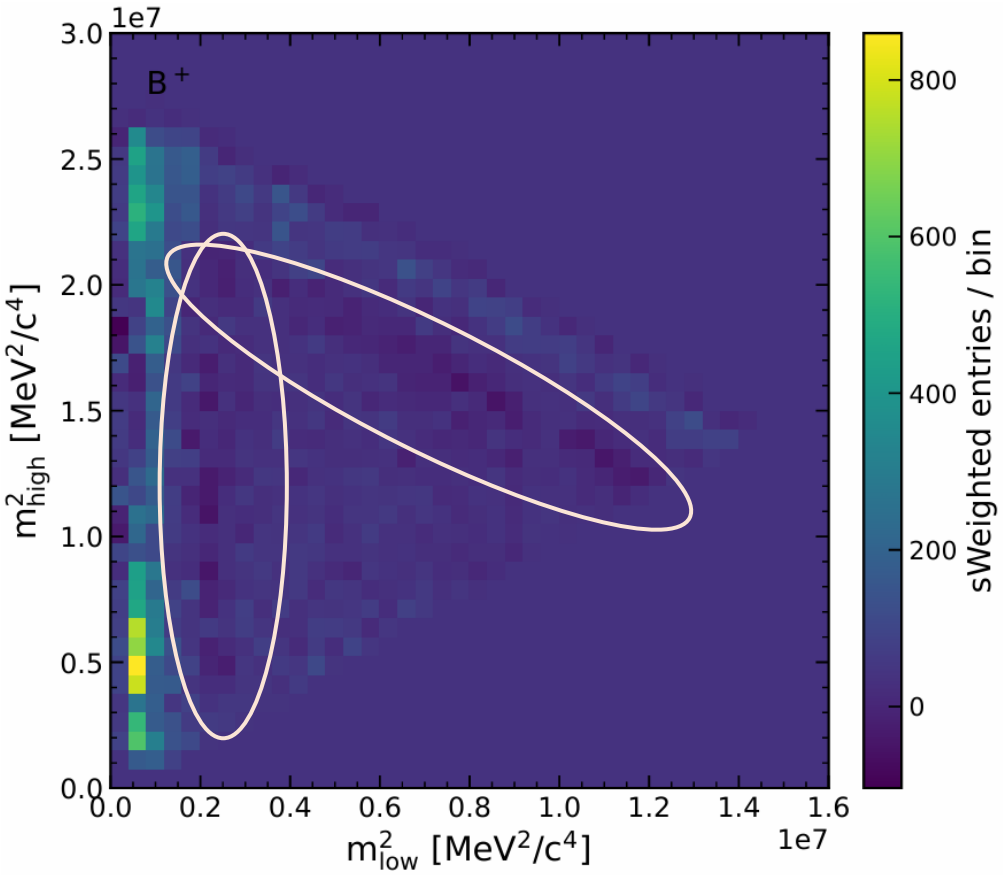
Correlation is weak → <u>proceed with caution</u>

sWeights result

Dalitz 1D Projection (Signal)



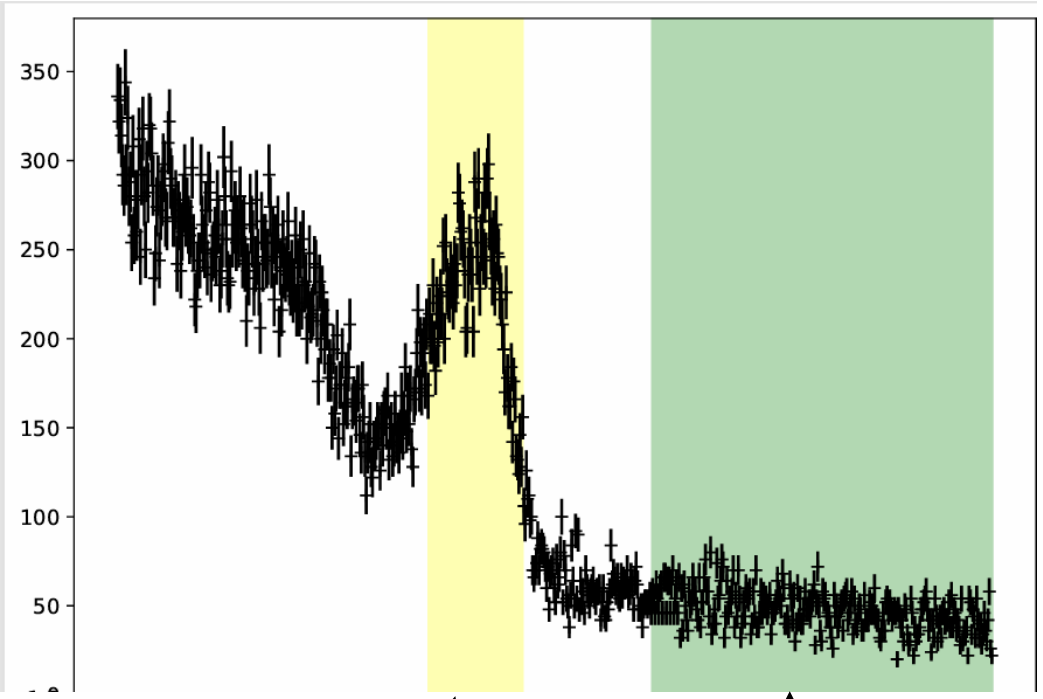
Signal-weighted Dalitz plot (B^+)



Overshoot creates **artificial structure** and negative weights

Clear resonance-like bands at low m^2_{low} as expected

Sideband subtraction methods



Selected signal region

Selected sideband region

Overview

1. Identify the combinatorial background only region(sideband) and signal region without 4body decay by global fitting

2. Model the background in Dalitz plot using the sideband data.

3. Scale the modelled background and subtract it from signal region data to evaluate true signals

Assumption

statistical independence between the Dalitz plot and the invariant mass. This is justified by the unstructured nature of combinatorial background events.

Combinatorial Background Modelling

Basis Functions (2D Chebyshev)

$$\hat{B}\left(x', y'; \{c_{ij}\}\right) = \sum_{i}^{N_x} \sum_{j}^{N_y} c_{ij}\, T_i(x') T_j\{y')$$

Domain Transformation

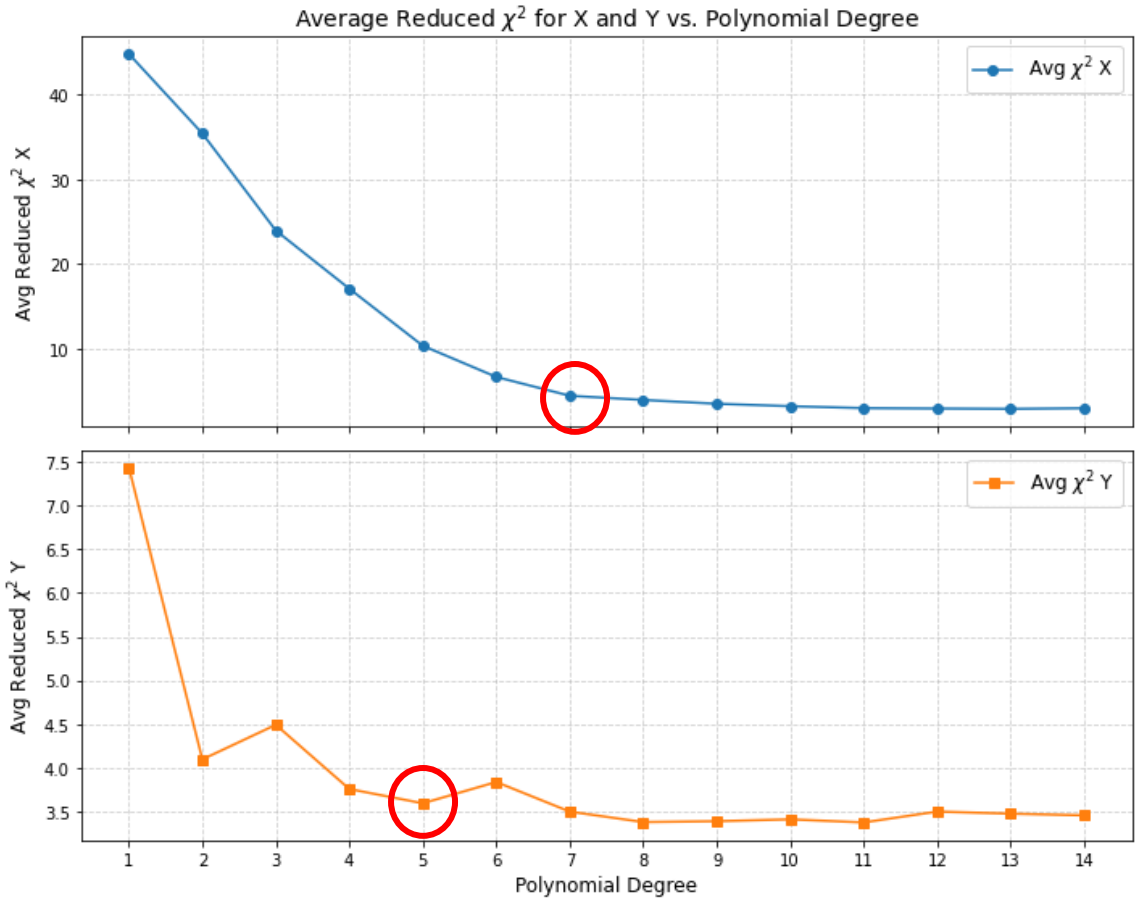
$$x' = \frac{2(x - x_{min})}{x_{max} - x_{min}} - 1$$

Minimisation(Ordinary Least Square)

$$S(c) = \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \left(B_{ij} - \hat{B}_{ij}\right)^2$$

- **Orthogonality & Stability**: Using Chebyshev polynomials minimizes correlations between the coefficients (c_{ij}).
- **Domain Mapping**: Mapping data to to $x' \in [-1,1]$ is essential for Chebyshev polynomials.
- **Independent Modelling**: It accounts for potential asymmetries across the 2D phase space.
- **Hyperparameter Tuning:** The maximum degree of Chebyshev polynomials must be determined.

Hyperparameter Tuning

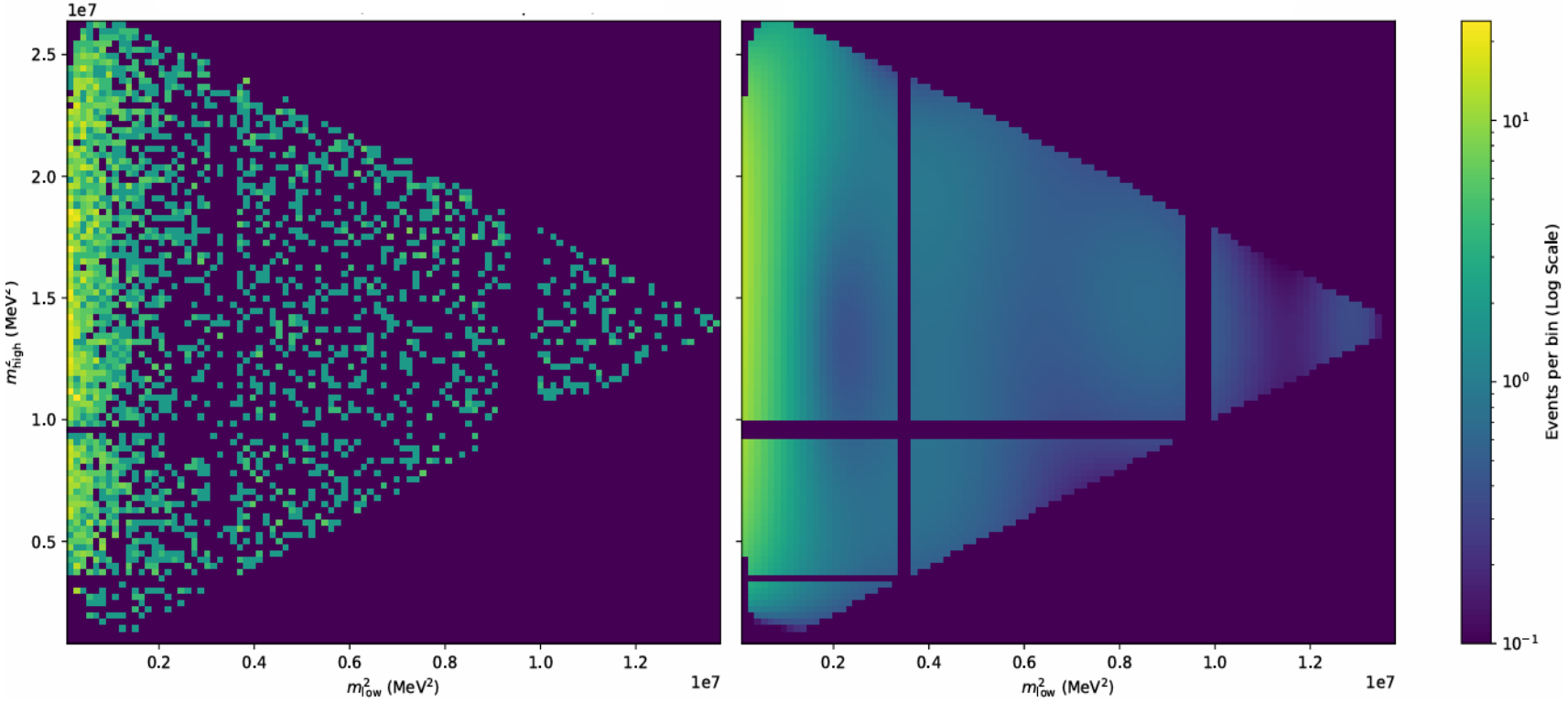


- **Optimisation**: Record reduced chi2 on the 1D projections of each axis.
- **Model Selection**: The maximum degrees are determined as $(N_x, N_y) = (7, 5)$. This avoids overfitting

Sideband modelling results 2D

B^- Binned Sideband Data

B^- Modelled Sideband Data



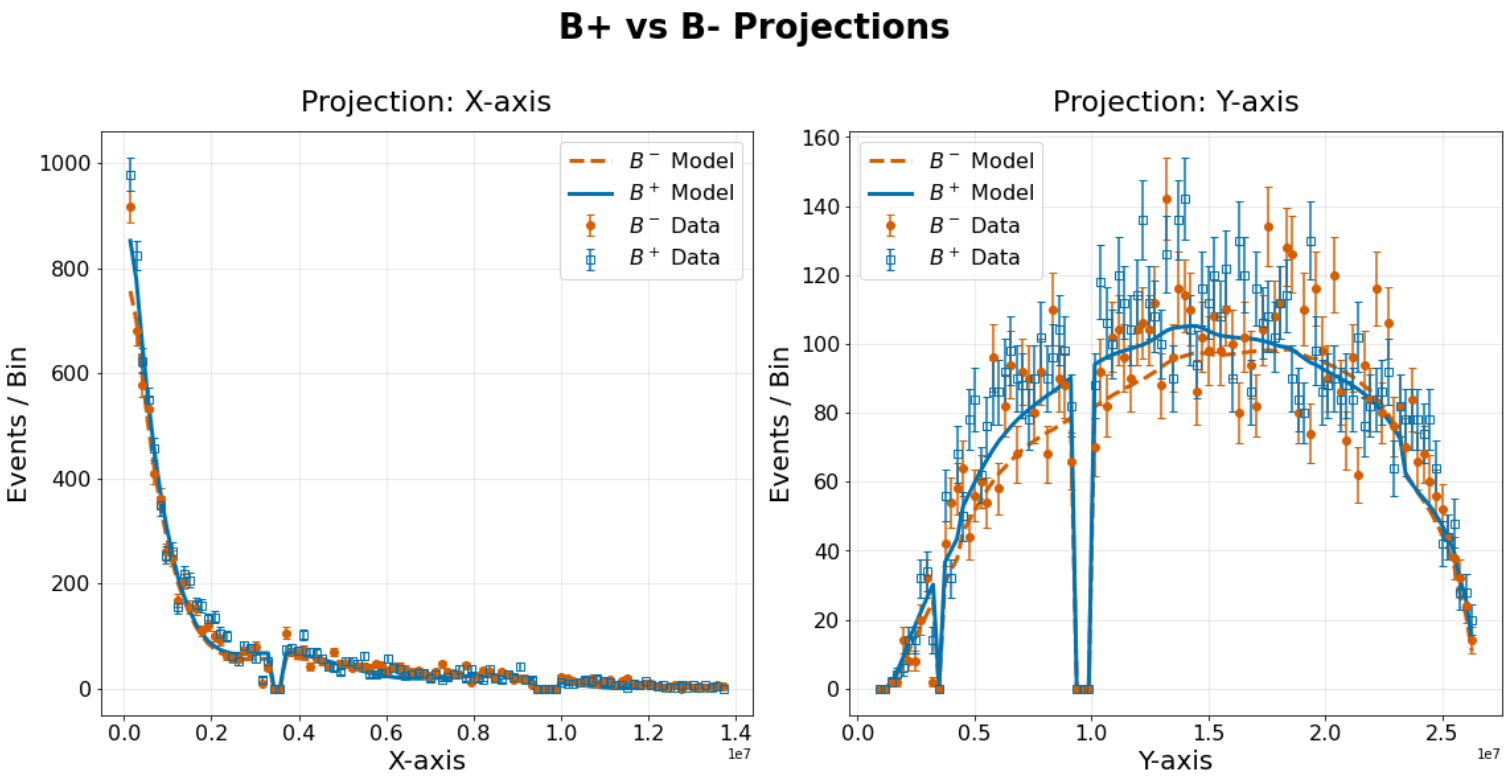
100 x 100 bins

- **Kinematic Boundaries**: $\max(s_{23}) = m_2^2 + m_3^2 + 2E_2^*E_3^* + 2p_2^*p_3^*$

(*: variable in the reference frame of 1)

- **Veto Handling**: Regions corresponding to $D^0 \to 3\pi$ & J/ψ resonances were manually masked.

Sideband modelling results 1D



- **Fit Quality**: Reduced χ^2 ranges from 2.0 to 6.0.
- **Observation**: Non-identical shapes between $B^{\pm}$ justifies independent modelling to account for the asymmetry

Notes on error estimation

$$\widehat{\mathrm{B}}(\boldsymbol{r}_i;\boldsymbol{\theta}) = \sum_j^m A_{ij}\,\theta_j$$

$\widehat{\mathrm{B}}$ is the estimated value. A is the design matrix. $\boldsymbol{\theta}$ is flattened coefficient (c_{ij}) vector

$$U = (A^T V^{-1} A)^{-1} = \widehat{\sigma}^2 (A^T A)^{-1}$$

U is the covariance matrix of estimated parameters. V is covariance matrix of raw data. For estimation, it's replaced with $V = \hat{\sigma}^2 \times I$

$$\widehat{\sigma}^2 = \frac{\sum\left(\mathrm{B} - \widehat{\mathrm{B}}\right)^2}{N - m}$$

$\hat{\sigma}^2$ is mean error squared (MSE). N is the number of data and m is the number of parameters.

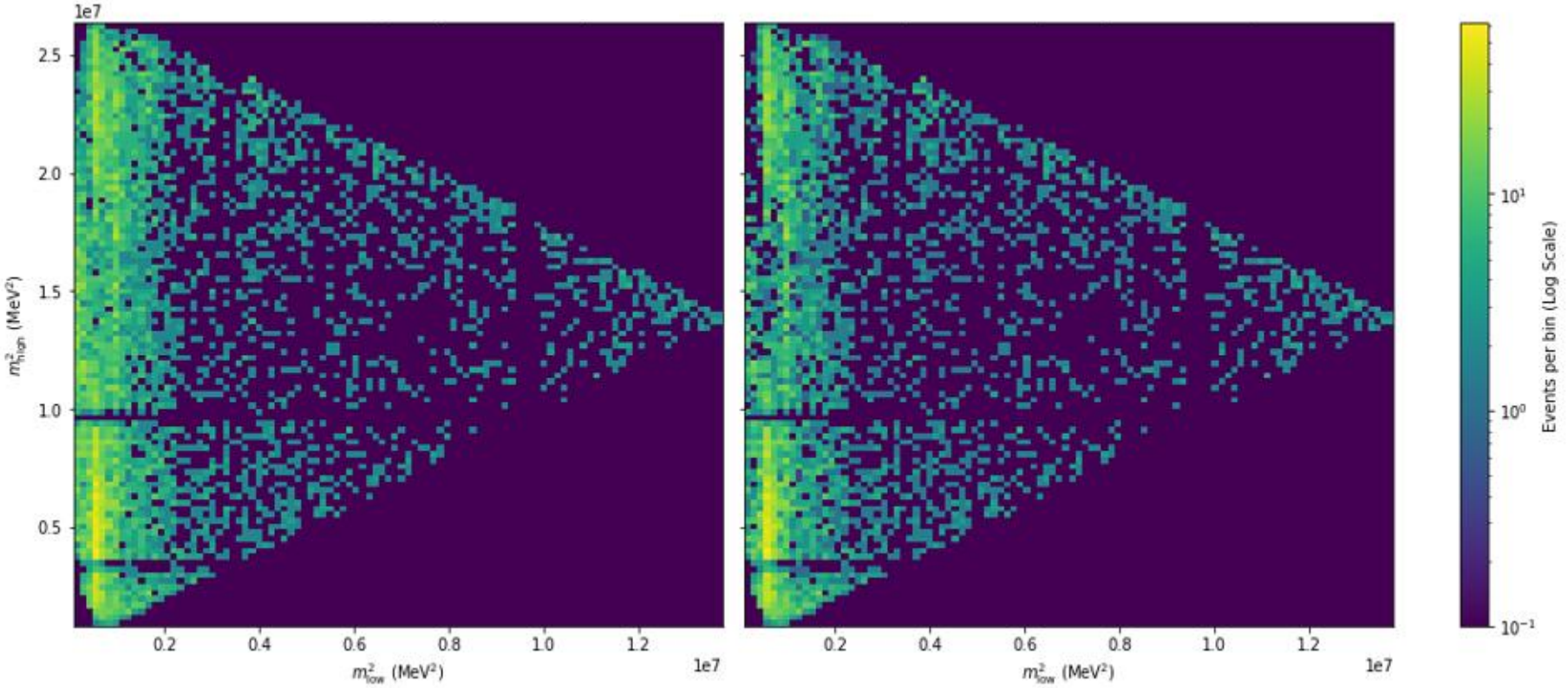
Ref [6]

$$Var\left(\hat{B}_i\right) = Cov\left(\widehat{\boldsymbol{B}}\right)_{ii} = (AUA^T)_{ii}$$

We can come back if detailed discussion is needed

Sideband subtraction results

B^- Signal before subtraction **B^- Signal after subtraction**



100 x 100 bins

Total Error:
Raw data Poisson noise + scaling factor uncertainty + background model's error

Comparison of methods

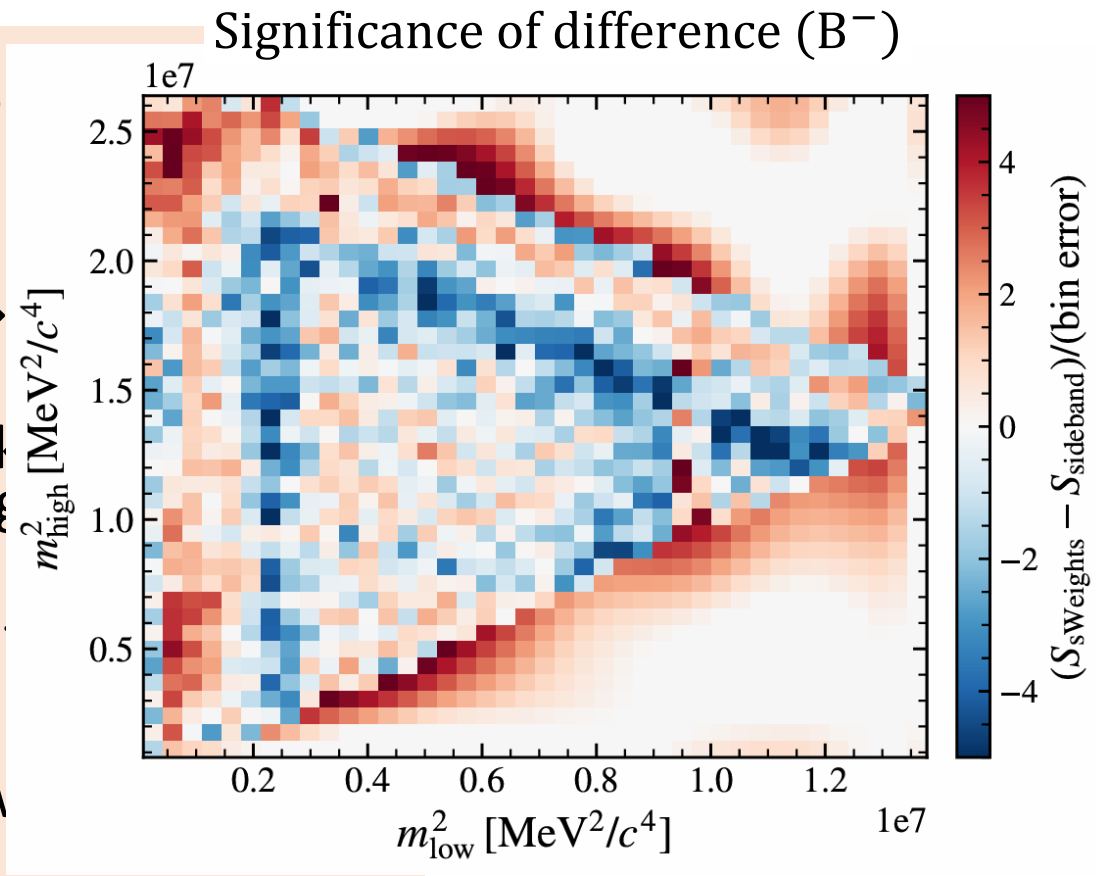
sWeights	Sideband subtraction
✓ Full dataset ✓ Naturally accounts for overlap	✓ Conceptually straightforward ✓ Does not rely on ARGUS fit
× Relies heavily on ARGUS fit × Can produce negative weights	× Uses only portion of data
Both require mass-Dalitz independence. • sWeights: independence across full distribution • Sideband subtraction: independence for combinatorial background only	

The two methods are complementary, with different sources of systematic uncertainty

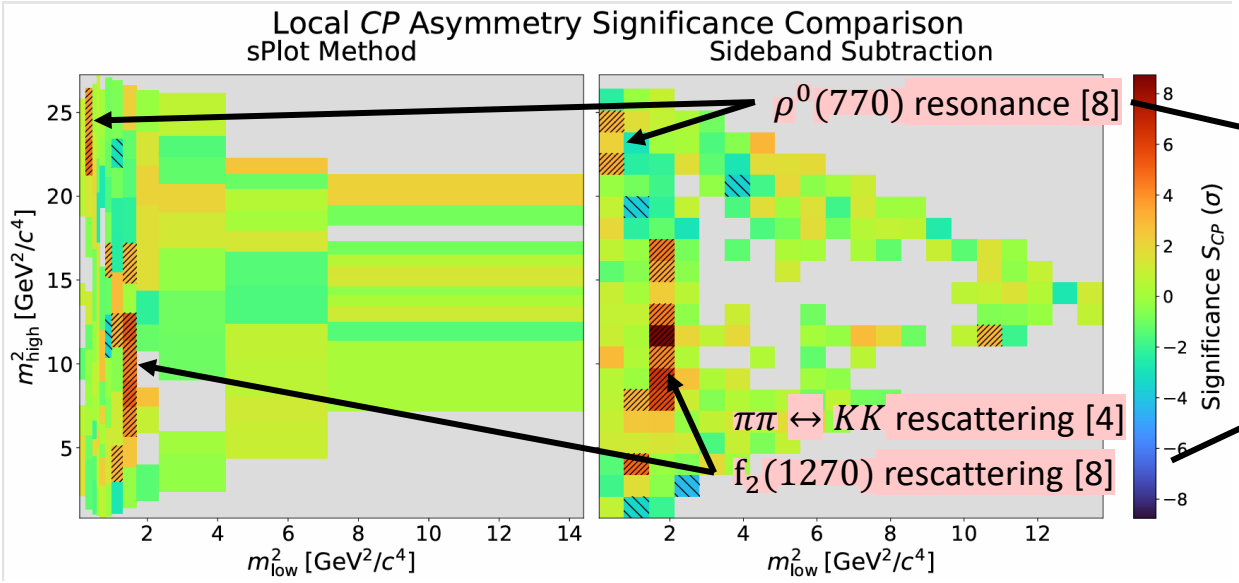
Comparison of subtractions

- Significant deviation in so
- Negative deviation (blue) overshoot
- Positive deviation (red) → resonance structure
- Edge differences (red) → k boundary + real modelling
- No charge-dependent bia asymmetry has smaller di

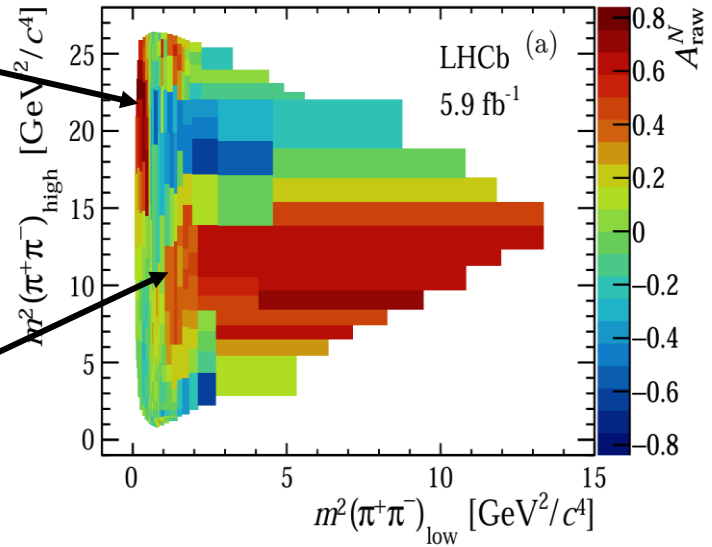
Overall: Two methods show deviation



Local CP Asymmetry Result



Previous study(Sweights)[4]



- Coarse adaptive binning chosen, with each bin having equal variance for Sweights methods.
- Consistent results obtained with sideband subtraction
- High-significance regions are spatially connected
- Absolute asymmetry reach ~1 >10x global asymmetry
- **Clear localised asymmetry regions**

5. Systematic Uncertainties and Conclusions

Systematic uncertainties

- Mass fit model
  - Model choice affects N^+, N^-, biases asymmetry (and sWeights)
  - Tested by varying signal shape (crystal ball, gaussian, bifurcated gaussian) + background model (Chebyshev, exponential)
  - Estimating uncertainty as half the range: $\sim 12\%$
- Efficiency correction
  - Detector efficiency different for $B^\pm$ and/or across Dalitz plot
  - LHCb estimated using simulation + binning variations: $\sim 2.3\%$ [4]
- Production asymmetry
  - pp collisions produce $B^\pm$ at slightly different rates
  - LHCb estimates using a control channel: $\sim 1.4\%$ [4]

Total systematic: $\sim 12\%$ (small)

Statistical uncertainty: $\sim 43\%$

Conclusions

- Measured CP asymmetry in $B^{\pm} \to \pi^{\pm}\pi^{+}\pi^{-}$ using LHCb data
- Global asymmetry: $(0.07 \pm 0.03) \to$ **2.3σ significance**
  - Consistent with LHCb results
- Strong local CP violation observed
  - Significances >8σ in specific Dalitz regions
  - Indicates phase-space dependent asymmetry from interfering resonances
- Method validation
  - sWeights and sideband subtraction give consistent results
- Limitations
  - Dominated by statistical uncertainty
  - Systematic uncertainty dominated by mass fit model dependence

    → improve selection process and fitting procedure with gaussian convolution.

References

[1] M. Kobayashi and T. Maskawa, CP Violation in the Renormalizable Theory of Weak I

[2] S. Navas et al. (Particle Data Group), Phys. Rev. D 110, 030001 (2024) and 2025 update.

[3] LHCb Collaboration, R. Aaij *et al.*, *LHCb detector performance*, Int. J. Mod. Phys. A **30** (2015) 1530022, doi:10.1142/S0217751X15300227.

[4] R. Aaij et al. (LHCb Collaboration), "Direct CP violation in charmless three-body decays of B+/- mesons," Phys. Rev. D 108, 012008 (2023).

[5] M. Mikhasenko *et al.*, *Dalitz-plot decomposition for three-body decays*, Phys. Rev. D **101** (2020) 034033, doi:10.1103/PhysRevD.101.034033.

[6] Cowan, G. (1998). *Statistical data analysis*. Oxford University Press.

[7] R. Aaij et al. (LHCb Collaboration), "Amplitude analysis of the $B^+ \to \pi^+\pi^+\pi^-$ decay," Phys. Rev. D 101, 012006 (2020). DOI: 10.1103/PhysRevD.101.012006

[8] LHCb Collaboration, R. Aaij et al., "Observation of several sources of CP violation in $B^+ \to \pi^+\pi^+\pi^-$ decays", Phys. Rev. Lett. 124, 031801 (2020) (arXiv:1909.05211).